

Program 8

STEP 1 : write a node.js app http server

```
(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab8]
$ ls
nodejs-configmap.yaml  nodejs-pod.yaml  node_modules  package.json  package-lock.json  server.js

(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab8]
$ cat server.js
const express = require("express");
const port = 3000;
const app = express();

app.get("/", (req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

app.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

STEP 2 : Config Map (nodejs-config.yaml)

```
(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab8]
$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

STEP 3 : Pod yaml

```
(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab8]
```

```
$ cat nodejs-pod.yaml
```

```
# apiVersion: v1
# kind: Pod
# metadata:
#   name: program-8
#   labels:
#     app: nodejs
# spec:
#   containers:
#     - name: nodejs
#       image: node:latest
#       imagePullPolicy: IfNotPresent
#       command: ["node", "/usr/src/app/server.js"]
#       volumeMounts:
#         - name: app-code
#           mountPath: /usr/src/app
#       ports:
#         - containerPort: 3000
#   volumes:
#     - name: app-code
#       configMap:
#         name: nodejs-app-config
```

```
apiVersion: v1
kind: Pod
metadata:
  name: program-8
spec:
  containers:
    - name: nodejs
      image: node:latest
      imagePullPolicy: IfNotPresent
      command: ["node"]
      args: ["/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /app
      ports:
        - containerPort: 3000

  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

STEP 4 : Apply the yaml files

STEP 5 : check pod status

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl apply -f nodejs-pod.yaml

pod/program-8 created

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-82hbb   1/1     Running   0           7m39s
nginx-deployment-6f9664446b-nd85k   1/1     Running   0           7m39s
program-8                            1/1     Running   0           8s

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl get configs
error: the server doesn't have a resource type "configs"

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl get configmaps
NAME                                DATA   AGE
kube-root-ca.crt                   1       25d
nodejs-app-config                   1       25d

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl logs nodejs-pod
error: error from server (NotFound): pods "nodejs-pod" not found in namespace "default"

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl logs program-8
Server running at port 3000

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl port-forward pod/nodejs-pod 3000:3000
Error from server (NotFound): pods "nodejs-pod" not found

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl port-forward pod/program8 3000:3000
Error from server (NotFound): pods "program8" not found

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab8]
$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 → 3000
Forwarding from [::1]:3000 → 3000
```

STEP 6 : Access the node js app

